Qi Le

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AI / ML Research Intern
Efficient and Trustworthy Language Models

Ph.D. student researching efficient and trustworthy language models. First-author work on dynamic LLM pruning and federated learning; experience in model compression, adaptation, alignment, and GPU-based evaluation.

Education

- **University of Minnesota, Twin Cities** | *Ph.D. Computer Science* — Sep 2022 – Present
- **University of Minnesota, Twin Cities** | *M.S. Data Science* — Aug 2020 – May 2022
- **Rensselaer Polytechnic Institute** | *B.S. Chemical Engineering, Minor in CS* — Aug 2016 – May 2020

Technical Skills

- **Research stack:** Python, PyTorch, Hugging Face Transformers / Datasets, NumPy; GPU inference benchmarking.
- **Methods:** LLM pruning and merging, KV-cache compression, PEFT, alignment, federated learning.

Research Experience

- **Research Assistant**
  University of Minnesota, Twin Cities — May 2021 – Present
  Advised by Prof. Ali Anwar, Prof. Jie Ding, and Dr. Enmao Diao.

Probe Pruning: Dynamic LLM Inference — *First author | ICLR 2025*

- Developed batch-adaptive structured pruning using lightweight model probes and historical activations to select attention heads and MLP channels, without fine-tuning; evaluated on LLaMA-2/3 and OPT.
- Achieved **1.46× attention-block and 1.30× MLP-block speedups** at 40% pruning on LLaMA-2-7B / WikiText2 (NVIDIA A100). Released open-source code.

Model Merging and KV-Cache Compression — *First author | ICLR 2027 submissions*

- Researching model merging through iterative parameter replacement (**OBMerge**) and KV-cache eviction with output-projected, OBS-inspired compensation (**ProjCache**).

Structured Pruning for Language and Reasoning Models — *Coauthor | ACL / DAC 2026*

- Contributed to GISP, an iterative, task-aligned global pruning framework. At 50% pruning, obtained **1.70× prefill throughput and 45% lower inference memory** on LLaMA-2-7B (A6000, batch 1). **ACL Outstanding Paper.**
- Contributed to RESP, which regenerates reasoning traces during progressive pruning; retained **81.27% GSM8K accuracy at 40% pruning** on Qwen3-8B, versus 14.40% for GISP in the same evaluation.

LLM Alignment and Content Safety — *Coauthor | ICLR / NAACL 2025*

- Contributed to MAP, using constrained optimization to align language models with user-defined human-value targets; evaluated decoding- and PPO-based alignment across six reward dimensions. **ICLR Oral.**
- Contributed to AID, combining eight safety detectors with input-dependent weights; achieved a **0.12 median absolute AUC gain** over the best individual detector across benchmark detection tasks.

Federated Learning and Efficient Model Adaptation — *First / second author | 2024*

- Developed DynamicFL to allocate client communication frequencies under resource budgets. Improved CIFAR-10 accuracy from **44.4% to 49.8%** over random client selection at a matched communication schedule (CNN, one class per client). **Best Student Paper.**
- Contributed to ColA, offloading adapter-gradient computation to separate devices. Llama-2 adaptation used **42.8 GB on the primary GPU** at batch 32, while the corresponding LoRA setup exceeded 48 GB (Dolly, all projections; extra offload-device memory).

Publications

- **Q. Le**, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar, **OBMerge: Optimal Brain Merge via Iterative Parameter Replacement**, *Submitted to ICLR 2027*
- **Q. Le**, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar, **ProjCache: Output-Projected KV Cache Eviction with OBS-Inspired Compensation**, *Submitted to ICLR 2027*
- Z. Wang, E. Diao, **Q. Le**, et al., **Think Before You Prune: Self-Reflective Structured Pruning for Reasoning Language Models**, *DAC 2026*
- Z. Wang, E. Diao, **Q. Le**, et al., **From Local to Global: Revisiting Structured Pruning Paradigms for Large Language Models**, *ACL 2026* Outstanding Paper
- J. Tang, X. Wang, **Q. Le**, et al., **AHE: Adaptive Homomorphic Encryption for Customizable Privacy in Heterogeneous Federated Learning**, *IEEE INFOCOM 2026*
- X. Wang, J. Du, A. A. Khan, **Q. Le**, et al., **Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models**, *NeurIPS 2025* Spotlight
- **Q. Le**, E. Diao, Z. Wang, et al., **Probe Pruning: Accelerating LLMs through Dynamic Pruning via Model-Probing**, *ICLR 2025*
- X. Wang, **Q. Le**, A. Ahmed, et al., **MAP: Multi-Human-Value Alignment Palette**, *ICLR 2025* Oral
- X. Wang, E. Diao, **Q. Le**, et al., **AID: Adaptive Integration of Detectors for Safe AI with Language Models**, *NAACL 2025*
- A. F. Khan, X. Wang, **Q. Le**, et al., **IP-FL: Incentive-Driven Personalization in Federated Learning**, *IEEE IPDPS 2025*
- **Q. Le**, E. Diao, X. Wang, et al., **DynamicFL: Federated Learning with Dynamic Communication Resource Allocation**, *IEEE BigData 2024* Best Student Paper
- X. Wang, **Q. Le**, A. F. Khan, et al., **ICL: An Incentivized Collaborative Learning Framework**, *IEEE BigData 2024*
- E. Diao, **Q. Le**, S. Wu, et al., **ColA: Collaborative Adaptation with Gradient Learning**, *arXiv:2404.13844, preprint, 2024*
- **Q. Le**, E. Diao, X. Wang, et al., **Personalized Federated Recommender Systems with Private and Partially Federated AutoEncoders**, *Asilomar 2022*

Honors and Awards

- **Outstanding Paper Award** | ACL 2026
- **Machine Learning Systems Fellowship** | Amazon 2025–2026
- **Best Student Paper Award** | IEEE BigData 2024

Teaching and Professional Service

- **Teaching Assistant**
  University of Minnesota, Twin Cities Mar 2021 – Present
  - Business Analytics; Algorithms and Data Structures; Introduction to Algorithms and Data Structures.

- **Reviewer**: ICLR 2027; NeurIPS 2025; Neural Processing Letters and IEEE BigData 2023; Asilomar 2022.
- **Co-reviewer (assisting faculty)**: USENIX ATC; USENIX FAST; IEEE Transactions on Parallel and Distributed Systems (TPDS).